

Description of different Components of Single bus Organization

Mainly there are 9 components associated with single bus Organization. they are:

1. program Counter

It is a register that manages the address of inst. to be executed next

2. R_0 --- R_{n-1}

This are GPR (Generally purpose register) and holds temporary while performing different operation

3. MAR

4. MDR

5. IR

6. ALU

7. Y, Z and TEMP

This are temporary registers that can be read or written into more than once in a single instruction (If needed)

8. MUX

It perform selection process on 2 or more incoming signal using data from input to combine the incoming signal on one or more o/p lines (Basically one o/p line)

9. SELECT

It is ranging from 0-1 and it is use for activating the MUX.